Data Attribution and Sources

Primary Data

• V-Dem (Varieties of Democracy) Country-Year Full+Others, version 15 (2024 data release). Indicators used: v2cltrnslw, v2juhcind, v2juncind, v2clkill, v2cltort, v2xcl_acjst, v2x_freexp, v2x_clphy, v2clrspct.

• World Bank World Development Indicators (WDI) via API — GDP per capita (current US$), population, prevalence of undernourishment, and poverty headcount at $2.15 (2017 PPP). Values are the most recent published as of atlas production (labeled 2026 baseline).

• United Nations Member States list (193 sovereign states) for country scope and ISO3 codes.

Source Framework

The RTLDI equation, the nine RTLP indicators, and the nested causal interpretation are taken from:
Sid J.A. Hubbard, Causality and Attraction: A Continuum of Steady States (Version 3, May 2026), DOI: 10.5281/zenodo.19468550. The operational crosswalk and binarization thresholds used here are documented in docs/indicator_crosswalk.md of this project.

This PDF was generated programmatically from the open RTLDI ATLAS toolkit. No proprietary or restricted data were used.

RTLDI Atlas Suite and Public Repository

This PDF atlas was produced by the RTLDI Atlas Suite, the open repository that contains the canonical data crosswalks, Python build pipeline, choropleth map generation, PDF ebook generator, and self-contained HTML atlas generator used for the public release.

Repository: https://github.com/SJAH9/rtldi-atlas

Users of this atlas are invited to download the RTLDI Atlas Suite and reproduce the published RTLP scores and RTLDI capital-exclusion projections directly from the public-release V-Dem and World Bank datasets. The suite is designed so NGOs, researchers, policymakers, analysts, and investors can rerun the atlas as new public data releases become available.

The same tools can also be used to generate an RTLDI Atlas from proprietary, internal, or field data. The implementation separates the conceptual nine-lever RTLDI model from the data crosswalk: if an organization has its own trusted country, regional, sectoral, or subnational indicators, it can map those fields into the required nine binary RTLP components and then use the existing suite to compute R, per-capita ΔG, total capital exclusions, lever-level attribution, regional summaries, maps, spreadsheets, PDF reports, and HTML outputs.

For public-data runs, use the documented V-Dem and World Bank workflow. For proprietary-data runs, implement a clear data crosswalk, preserve the nine RTLP component meanings, document all thresholds, and keep the output labels tied to the same canonical RTLDI equation. This keeps forks comparable while allowing local datasets to improve coverage, timeliness, or granularity.

The official public atlas has one canonical pipeline: src.build_atlas produces the master atlas table; src.generate_atlas_ebook.prepare_atlas_data() prepares the shared print/web data; and the PDF/HTML generators consume that same prepared data. Forks should preserve this single-pipeline discipline or clearly identify themselves as experimental derivatives.

Nested Causal Modelling

The atlas takes the RTLDI from the source document and extends it via population-weighted regression (25% cap) to quantify capital exclusions — the lost potential GDP where capital is excluded from the economy because the nine protections required for it to operate are not present. The nine levers are the conditions from the source modelling. Their descriptions and the scale of exclusions associated with each are given in the front-matter section. The 25 % cap is a practical limiter on projections; the levers are low-cost conditions whose activation can bring excluded capital back in.

Index of Terms

2026-fresh G0:
: The most recent published GDP per capita values used as the dynamic baseline for the 2026 edition (even when RTLP components are from 2024).

Access to Justice:
: Fair access to remedies for right-to-life violations

Arbitrary Detention:
: Core protection against state overreach (RTLP indicator #4); current implementation level contributes to associated capital exclusions in regions.

Attributable capital exclusions:
: The share of a country or region's total capital exclusions (lost potential GDP) that is directly traceable to the current implementation level of one specific protection (sum of $\eta/9 \times G_0 \times$ pop over countries where the protection is not fully present, with current $\eta \approx 0.30$ from population-weighted cross-section).

Binarization / threshold:
: The documented conversion of continuous V-Dem and World Bank values into the binary 0/1 scores for the nine indicators (e.g., ≥2.0 on 0-4 scales, ≤5 % undernourishment + ≤10 % poverty for #9).

Causality and Attraction:
: The source monograph (Hubbard 2026V3, Zenodo 19468550) that supplies the RTLDI equation, the nine RTLP indicators, the nested-enclosures paradigm, and the cartographic critique of distortion. This atlas extends the source RTLDI to quantify capital exclusions.

Choropleth:
: Thematic map in which countries or regions are shaded by enclosure strength R (0–1) using the RTLDI atlas scale; primary visual for the global, regional, and focus views.

Civilian Protection in Conflict Zones:
: Mechanisms protecting civilians in conflict areas

Country view:
: The country-only choropleth on each nation page, fitted to the national boundary and shaded by enclosure strength R without using the regional context map.

Crosswalk:
: The explicit mapping from the nine verbatim RTLP questions to observable V-Dem variables and World Bank statistics, with defined binarization thresholds.

ΔG (delta G):
: Annual per-capita capital exclusions (lost potential GDP) associated with the current level of the nine levers. Core output of the RTLDI equation.

Economic Intelligence (tool):
: The use of R scores, component breakdowns, and attributable capital exclusions figures as practical information for governments, businesses, and analysts to identify high-leverage institutional opportunities.

Enclosure (nested causal / human):
: The source document's central metaphor: the interdependent, layered structures of protection whose strength or weakness shapes higher-order economic and social steady states.

η (eta):
: The sensitivity coefficient (current data-driven default 0.30 from population-weighted 2026 UN cross-section). In the bounded model $\Delta G = \min(\eta(1 - R) \times G_0, 0.25 \times G_0)$; the cross-sectional premium per indicator, then capped so the nine RTLP indicators are never credited with more than 25% of observed G_0 after other determinants are given due weight. (Source document used a more conservative 0.05 structural value.)

Existence of Legal Protections:
: Provisions explicitly protecting the right to life

Freedom from Torture and Inhumane Treatment:
: Measures in place to prevent torture

Freedom of Expression and Whistleblower Protections:
: Expression and whistleblower rights upheld

G_0 (G-zero):
: Baseline GDP per capita (World Bank, current US$). The 'current' economic size against which capital exclusions are measured.

Independent Judiciary:
: Independent judiciary capable of upholding right-to-life laws

Law Enforcement Accountability:
: Law enforcement agencies accountable for unlawful killings

Member-nations table:
: The compact table on each regional summary page that lists every country in the region together with its R, G0, population, and capital exclusions, plus a bottom row of regional totals.

Mollweide:
: Equal-area pseudocylindrical map projection used for all choropleths (global, regional, and per-nation zooms) to ensure visual area corresponds to geographic reality.

Protection Against Arbitrary Detention:
: Arbitrary detention prohibited with legal recourse

R (lever activation score):
: Average activation level (0–1) of the nine Right to Life Deficit Index levers. 1 = full activation across all dimensions.

Right to Life:
: The overarching protected interest; the nine indicators operationalize equal and effective protection against arbitrary interruption.

RTLDI:
: Right to Life Deficit Index — the full framework from the source (Zenodo 10.5281/zenodo.19468550), extended here to quantify capital exclusions (lost potential GDP where capital is excluded because the nine protections necessary for it to operate are absent).

RTLDI atlas R scale:
: The shared perceptual color scale used for every enclosure-strength choropleth in the atlas: deep low-R tones through teal midrange values to warm high-R values.

Socioeconomic Conditions:
: Access to healthcare, food, and shelter adequate to sustain life

Steady state:
: A balanced biological, economic, or social condition that the framework argues is supported or undermined by the strength of the nine life protections and the resulting causal enclosures.

The Nine Levers:
: The nine binary institutional conditions (RTLDI) that are averaged to produce R. Low-cost, high-return switches identified through modelling for maximal prosperity.

Torture and Inhumane Treatment:
: RTLP indicator #5; one of the larger contributors to global capital exclusions, highlighting physical integrity protections.

Trend (3-year):
: The per-nation GDP-driven RTLDI plots (R held fixed at 2024 values) that illustrate recent dynamics in capital exclusions alongside the regional and global maps.

UN region / regional summary:
: One of the 22 geographic groupings used for aggregates, focused choropleths, member-nations tables (with R / G0 / population / capital exclusions + REGIONAL TOTAL row), and 9-lever breakdowns.

V-Dem:
: Varieties of Democracy project. Supplies the expert-coded governance and civil liberties indicators for eight of the nine RTLP components.

Credits and Acknowledgments

Conceptual Framework and Equations
Sid J.A. Hubbard — author of Causality and Attraction: A Continuum of Steady States (v3, 2026). All rights to the RTLDI equations, the nine RTLP indicators, and the nested causal paradigm remain with the author.

Data
• V-Dem Institute (https://www.v-dem.net) — V-Dem Country-Year dataset v15.
• World Bank Open Data (https://data.worldbank.org) — WDI via API.
• United Nations — list of 193 Member States.

Code and Atlas Production
RTLDI ATLAS project (2026). The open-source toolkit, crosswalk implementation, data pipeline, and this PDF generator are provided to make the framework usable by NGOs and researchers.

This work is released in the spirit of the original document: to map, with transparent data and reproducible math, the economic consequences of failing to protect the right to life equally.

For the full source code, 2026 data tables, and previous editions (2023, 2024), see the project repository. Users are encouraged to download the latest V-Dem and World Bank releases and re-run the pipeline for updated figures.